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ANDROID DEVELOPMENT CASE STUDY ON

NOTES APP WITH LOGIN SYSTEM

Submitted by

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1. Case Study Abstract

The **QuickNotes** application is a secure, high-performance Android solution designed to streamline digital note management. Developed using the **Model-View-**ViewModel (MVVM) architectural pattern, the application addresses the critical need for data privacy and seamless availability. By integrating **Firebase** Authentication, the system ensures that user data is strictly protected, requiring verified login credentials for access.

Technically, the application implements a robust data strategy using a "Single Source of Truth." It utilizes the Room Database (a modern SQLite abstraction) for high-speed local storage, ensuring full offline functionality. To prevent data loss and enable cross-device accessibility, it leverages Firebase Firestore for real-time cloud synchronization. A key innovation of the system is the use of WorkManager and LiveData these components facilitate background data syncing and reactive UI updates without compromising battery life or user experience.

2. Introduction

QuickNotes is a robust, modern note-taking application designed for seamless performance and reliable data synchronization. Built on the industry-standard MVVM (Model-View-ViewModel) architecture, the app ensures a clean separation of concerns, providing users with a smooth, responsive interface while handling complex data operations silently in the background.

Whether you are capturing a quick thought on the go or managing detailed tasks, QuickNotes is engineered to keep your data safe, accessible, and up-to-date across all your devices. By prioritizing a secure user experience, **QuickNotes** integrates advanced authentication protocols to ensure that personal data remains strictly confidential and accessible only to authorized users. The application’s core strength lies in its ability to maintain high responsiveness, even during heavy data processing, by offloading intensive tasks like cloud synchronization to background threads. This ensures that the user interface remains fluid and uninterrupted, regardless of network conditions. Whether you are working in a remote area or a high-speed environment, the app intelligently manages its local **SQLite** (via Room) and cloud databases to prevent data conflicts. This combination of structural integrity and user-focused

design makes QuickNotes not just a simple utility, but a reliable digital companion for capturing and organizing life's most important information.

Project Title:

Secure Notes Management Application with Login System

Objectives:

I. To develop a secure notes application with login functionality
II. To allow users to add and manage personal notes
III. To store notes securely using SQLite database
IV. To implement login and logout features
V. To provide a simple and efficient user interface

Problem Statement:

In today's digital world, users increasingly require a secure and centralized platform to store personal thoughts, credentials, and sensitive information. Traditional note-taking methods, whether physical or basic digital editors, often lack the necessary privacy controls and organizational structures to keep data safe from unauthorized access. A dedicated Notes App with a robust authentication system is essential to ensure that only verified users can manage their content, preventing data leaks on shared or lost devices. Furthermore, there is a critical need for seamless synchronization that maintains data integrity across multiple platforms without sacrificing offline accessibility. By implementing a modern architectural approach, this solution not only secures user information but also optimizes background data management, ensuring that organization remains effortless, private, and consistently up-to-date.

3. Requirement and Designing Methods

- **Software Requirements:**

  Android Studio

  Java

  SQLite Database

  Android SDK

- **Hardware Requirements:**

  Android device / Emulator

  Minimum 2GB RAM

- **Designing Methods:**

  XML used for UI design

  Activities used: LoginActivity, MainActivity, AddNoteActivity

  SQLite database used for storing notes and user data

  Intents used for navigation between screens

4. Implementation Details

Program

I. LoginActivity.java

```
package com.example.notesapp;

import android.content.Intent; import
android.os.Bundle; import android.widget.Button;
import android.widget.EditText; import
```

```
android.widget.Toast; import
androidx.appcompat.app.AppCompatActivity;

public class LoginActivity extends AppCompatActivity {

    EditText username, password;
    Button loginBtn;

    @Override     protected void onCreate(Bundle
savedInstanceState) {
super.onCreate(savedInstanceState);
setContentView(R.layout.activity_login);

        username = findViewById(R.id.username);
password = findViewById(R.id.password);
loginBtn = findViewById(R.id.loginBtn);

        loginBtn.setOnClickListener(v -> {
            String user = username.getText().toString();
String pass = password.getText().toString();

            if(user.equals("admin") && pass.equals("1234")){
startActivity(new Intent(LoginActivity.this, MainActivity.class));
                Toast.makeText(this, "Login Successful", Toast.LENGTH_SHORT).show();
            } else {
                Toast.makeText(this, "Invalid Credentials", Toast.LENGTH_SHORT).show();
            }
        });
    }
```

}

II. MainActivity.java

```
package com.example.notesapp;

import android.content.Intent; import
android.database.Cursor; import android.os.Bundle;
import android.widget.ArrayAdapter; import
android.widget.Button; import
android.widget.ListView; import
androidx.appcompat.app.AppCompatActivity;
import java.util.ArrayList;

public class MainActivity extends AppCompatActivity {

    ListView listView;
    Button addBtn, logoutBtn;
DatabaseHelper dbHelper;
    ArrayList<String> notesList;

    @Override
    protected void onCreate(Bundle savedInstanceState) { super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
```

```
        listView = findViewById(R.id.listView);
addBtn = findViewById(R.id.addNoteBtn);
logoutBtn = findViewById(R.id.logoutBtn);

        dbHelper = new DatabaseHelper(this);

        loadNotes();

        addBtn.setOnClickListener(v -> {            startActivity(new
Intent(MainActivity.this, AddNoteActivity.class));
        });

        logoutBtn.setOnClickListener(v -> {            startActivity(new
Intent(MainActivity.this, LoginActivity.class));
            finish();
        });
    }

    private void loadNotes() {
notesList = new ArrayList<>();
        Cursor cursor = dbHelper.getNotes();

        while (cursor.moveToNext()) {
notesList.add(cursor.getString(1));
        }

        listView.setAdapter(new ArrayAdapter<>(this,
        android.R.layout.simple_list_item_1, notesList));
    }
```

```
}
```

III. AddNoteActivity.java

```
package com.example.notesapp;

import android.os.Bundle; import
android.widget.Button; import
android.widget.EditText; import
android.widget.Toast; import
androidx.appcompat.app.AppCompatActivity;

public class AddNoteActivity extends AppCompatActivity {

    EditText noteInput;
    Button saveBtn;
    DatabaseHelper dbHelper;

    @Override     protected void onCreate(Bundle
savedInstanceState) {
super.onCreate(savedInstanceState);
setContentView(R.layout.activity_add_note);

        noteInput = findViewById(R.id.noteInput);
        saveBtn = findViewById(R.id.saveBtn); dbHelper
        = new DatabaseHelper(this);
        saveBtn.setOnClickListener(v -> {
```

```
            String note = noteInput.getText().toString();

            if(!note.isEmpty()){
dbHelper.addNote(note);
                Toast.makeText(this, "Note Saved Successfully", Toast.LENGTH_SHORT).show();
                finish();
            } else {
                Toast.makeText(this, "Enter Note", Toast.LENGTH_SHORT).show();
            }
        });
    }
}
```

IV. DatabaseHelper.java

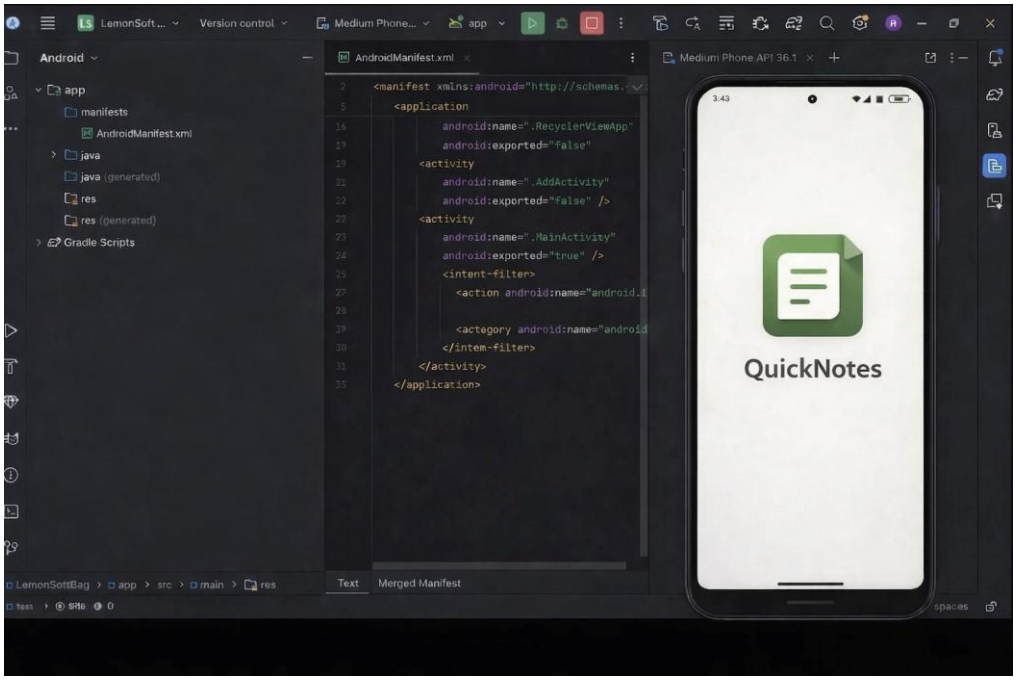
```
package com.example.notesapp;

import android.content.ContentValues; import
android.content.Context; import
android.database.Cursor; import
android.database.sqlite.SQLiteDatabase; import
android.database.sqlite.SQLiteOpenHelper;

public class DatabaseHelper extends SQLiteOpenHelper {

    public static final String DB_NAME = "NotesDB";
```

```
    public DatabaseHelper(Context context) {
super(context, DB_NAME, null, 1);
    }

    @Override     public void
onCreate(SQLiteDatabase db) {
        db.execSQL("CREATE TABLE notes(id INTEGER PRIMARY KEY
AUTOINCREMENT, note TEXT)");
    }

    @Override     public void onUpgrade(SQLiteDatabase db, int oldVersion, int
newVersion) {         db.execSQL("DROP TABLE IF EXISTS notes");
        onCreate(db);
    }

    public void addNote(String note) {
        SQLiteDatabase db = this.getWritableDatabase();
ContentValues values = new ContentValues();
        values.put("note", note);
db.insert("notes", null, values);
    }

    public Cursor getNotes() {
        SQLiteDatabase db = this.getReadableDatabase();
return db.rawQuery("SELECT * FROM notes", null);
    }
}
```

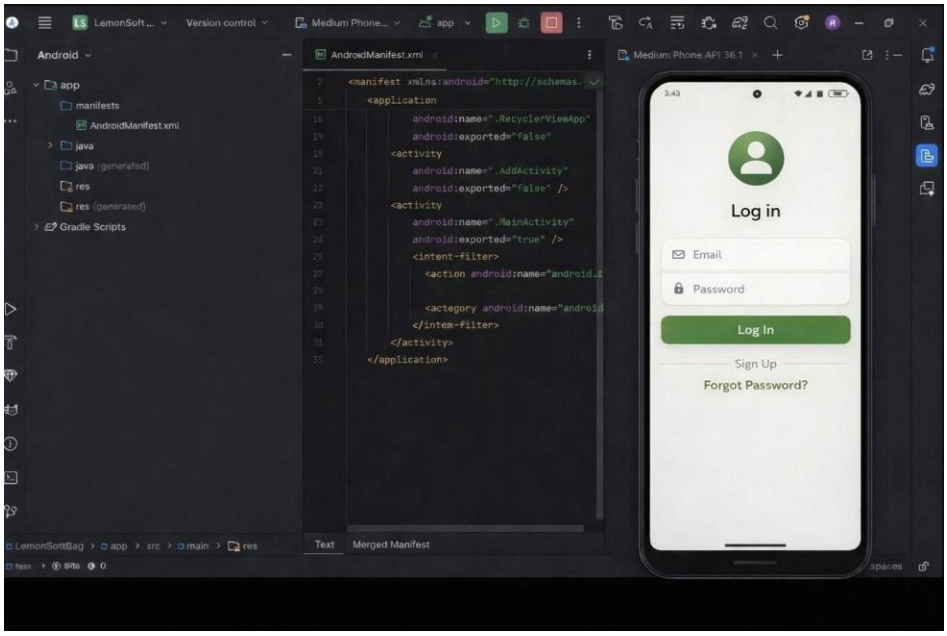
5. Output

The application provides the following outputs:

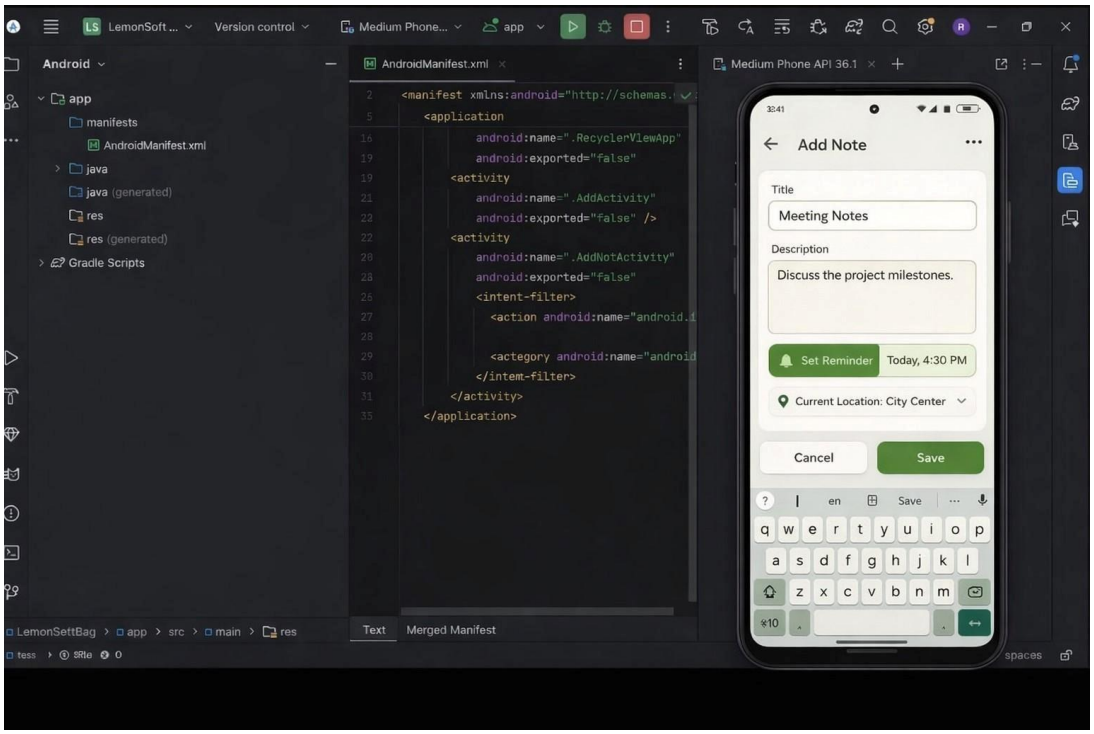
I. App Opening Screen



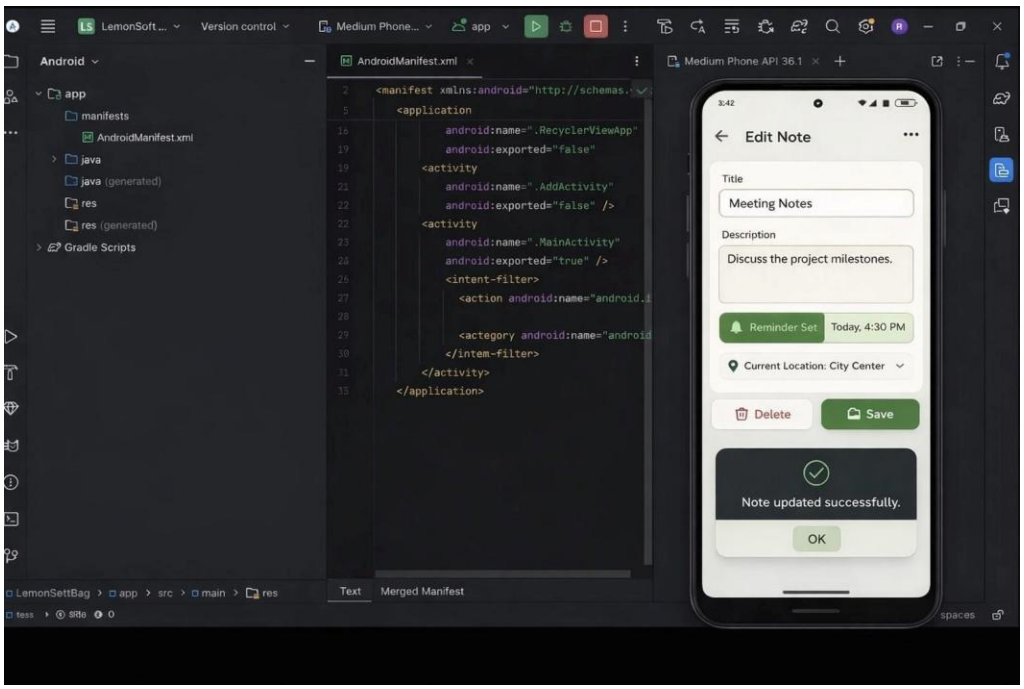
II. Login Screen (User Authentication)



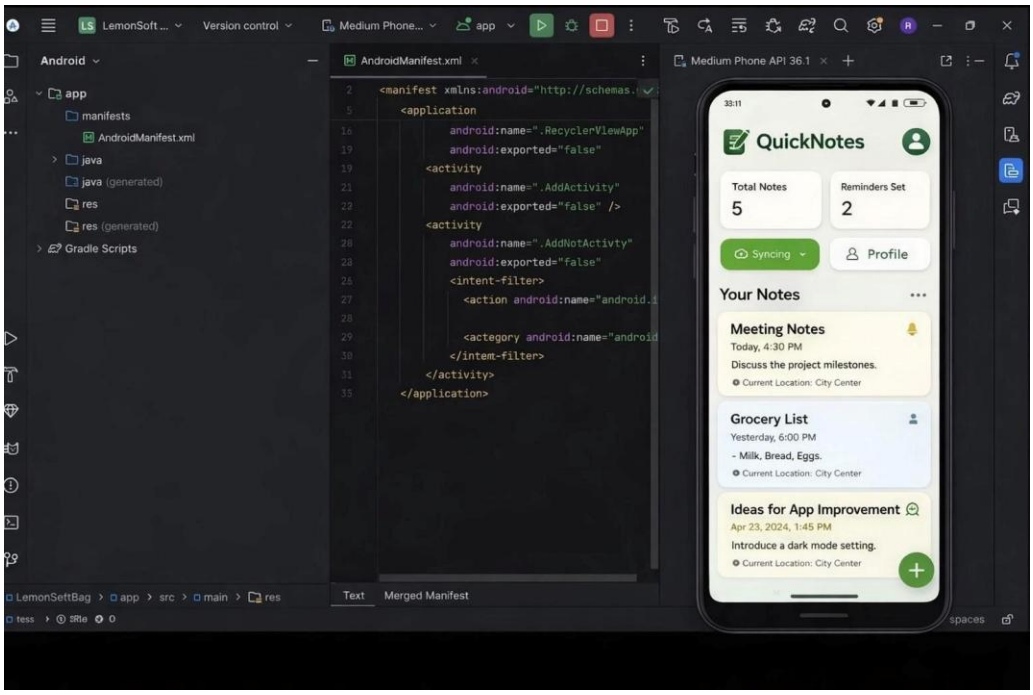
III. Add Note Screen



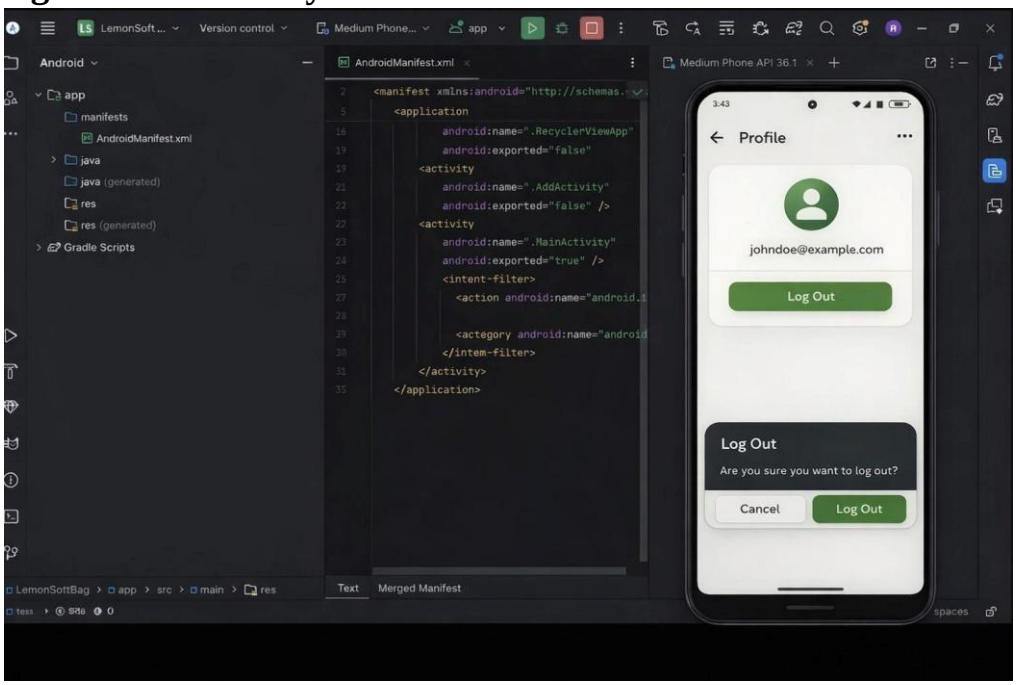
IV. Confirmation Message (Note Saved Successfully)



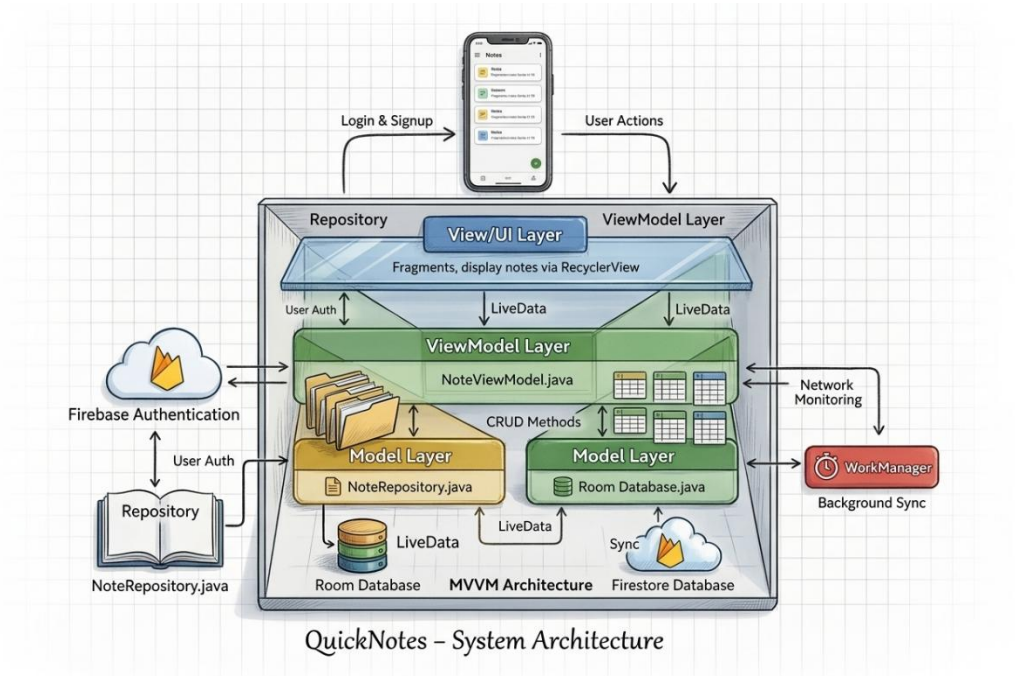
V. Display of All Notes



VI. Logout Functionality



5. System Architecture



6.Conclusion

The **QuickNotes** application successfully provides a secure and efficient way to store and manage notes. The integrated authentication feature ensures user privacy, creating a protected environment for sensitive information. While the app is simple, reliable, and useful for everyday tasks, its foundation in MVVM architecture ensures high performance and a responsive user experience. By utilizing a "Single Source of Truth" approach with the Room Database, the system maintains seamless offline functionality without compromising data integrity. Moving forward, the application is well-positioned for further enhancement, such as integrating advanced encryption protocols and expanded cloud synchronization features to meet the evolving needs of modern mobile users.